

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1170

Skaggs Island 138 kV
July 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1170. The Interconnection Customer proposes a 155.2 MW battery energy storage facility interconnecting at the Skaggs Island 138 kV substation in the Virginia area. The request was received on March 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 395 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$68,504,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Gaston Substation / Tecumseh Energy Ctr 345kV, the 32Nd St / Kekuk 138kV loads to 147.1% of its 237 MVA rating. Project DFAX: 14.0%.

The 32Nd St / Straits 138kV line is overloaded to 145.8% of its 236 MVA emergency rating for the loss of the Tecumseh Energy Ctr / Gaston 345kV. The project contributes 14.0% DFAX (21.7 MW) to this overload.

Straits / Dpc Elroy 138kV: 145.1% loading (234 MVA) following the outage of the Tecumseh Energy Center / Gaston 345kV; DFAX = 14.0%.

The Genev Border City / Keokuk 138kV is overloaded to 132.5% of its 193 MVA emergency rating for the loss of the Bear / Clover 345kV. The project contributes 6.0% DFAX (9.3 MW) to this overload.

Dewalt Substation / Old Hickory 138kV: 132.0% loading (275 MVA) following the outage of the N Tower Tap / Dpc Elroy 138kV; DFAX = 12.7%.

Geneva Border City / Dwalt 138kV: 123.2% loading (306 MVA) following the outage of the N Tower Tap / Dpc Elroy 138kV line; DFAX = 12.7%.

Geneva Border City / Wellsic 138kV: 120.1% loading (315 MVA) following the outage of the Tecumseh Energy Ctr / Gaston 345kV; DFAX = 6.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Keokuk Substation / 32Nd St 138kV Line	Rebuild line with higher-capacity conductor	\$7,162,000	100% (\$7,162,000)
32Nd St / Straits 138kV line	Rebuild line with higher-capacity conductor	\$8,348,000	100% (\$8,348,000)
Straits / Dpc Elroy 138kV	Rebuild line with higher-capacity conductor	\$16,259,000	100% (\$16,259,000)
Genev Border City / Keokuk 138kV	Rebuild line with higher-capacity conductor	\$3,955,000	100% (\$3,955,000)
Dewalt Substation / Old Hickory 138kV	Rebuild line with higher-capacity conductor	\$4,897,000	100% (\$4,897,000)
Geneva Border City / Dewalt 138kV	Rebuild line with higher-capacity conductor	\$18,878,000	100% (\$18,878,000)
Geneva Border City / Wellsic 138kV	Rebuild line with higher-capacity conductor	\$9,005,000	100% (\$9,005,000)
	Total allocated to project		\$68,504,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.